Dijkstra's Algorithm - Basics

Dijkstra's algorithm finds the shortest paths from a source node in a weighted graph. It works by greedily settling the closest unsettled node, again and again, until all reachable nodes are settled. Time complexity with a binary heap is O((V + E) log V). Page 1 introduces the greedy idea.

Dijkstra vs Negative Weights - Bellman-Ford

Dijkstra's algorithm does NOT support negative edge weights. Once it finalizes a node it never revisits it, so a later shortcut through a negative edge is missed and the answer is wrong. Bellman-Ford DOES handle negative edge weights (but not negative cycles) by relaxing every edge repeatedly, V-1 times. Use Bellman-Ford when any weight can be negative. Page 2 is the key exam fact.

Worked Example

Example: A -> B cost 4, A -> C cost 2, C -> B cost -3. Dijkstra from A settles B at 4 and stops, missing the better path A -> C -> B with total cost -1. Bellman-Ford relaxes all edges and finds -1. Rule of thumb: non-negative weights use Dijkstra; negative weights use Bellman-Ford. Page 3.